# Hands-on Workshop: Multiple particle tracking microrheology in planktonic systems

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## Learning objectives

In this workshop you will:

- Understand the general principles of multiple particle tracking microrheology (MPTM)
- Explore conceptual differences between MPTM and bulk rheology
- Use a field viscometer (µVISC) to measure bulk viscosity
- Prepare samples for microrheology experiments
- Acquire videos of microspheres undergoing Brownian motion
- Track beads and analyse trajectories to estimate rheological properties

## General Workflow

The workshop is organized in three activities (see below) of increased complexity. Let’s see how far we can go!

In all activities we will use the same basic steps of data collection and analyses. Here is a brief description of the main procedures.

### µrheology

To conduct microrheology we will use a technique called multiple particle tracking microrheology (MPTM), emplying a hand-made fluorescence microscope and a camera to track beads suspended in our sample. For details on how to conduct MPTM refer to the accompanying document (‘ViscosityMaps_protocol.pdf’). In general, the main steps to obtain and analyze videos for microrheology are:

- Seed 100µL of sample with 4 µL of the beads suspension.
- Mount a slide carefully using an adhesive spacer
- Acquire videos (~1 min, 20 fps) using the camera software (µeye)

### Data analyses

We will do the data analyses will be make with MATLAB, using the accompanying live script (‘MOB_MPTM.m’)

## µVISC operation

- Load your sample in a syringe
  (If sample contains >10 µm particles, filter)
- Run measurement X3  and record viscosity and temperature.
- Clean between samples by running mQ water.
- Remember to recalculate viscosity values using the calibration provided.

## ACTIVITIES

## Activity 1: Bulk vs Microrheology

**Objective**: We will compare viscosity measurements conducted with microrheology and bulk rheology on Newtonian fluids of different Ficoll concentration.

**Steps**:

1. Measure freshwater and at least 2 Ficoll solutions at different concentrations using the µVisc field viscometer.
2. Perform microrheology on the same solutions
   1. Prepare slides with Ficoll solutions
   2. Take 1 minute videos of the beads moving, at 20 fps frame rate.
   3. Analyse them with MATLAB to obtain mean viscosities
3. Create a scatterplot of the two kinds of measurements and fit a model

**Discussion points:**

- How do the two measurement approaches compare?
- What are the main sources of error?
- Can you estimate the temperature at the slide during measurements?

## Activity 2: Microrheology of a diatom Culture

**Objective**: We will map viscosity around a diatom cell or within an aggregate.

**Steps**:

1. Prepare slides with a sample from the diatom culture, (as in the activity 1)
2. Take 1 minute videos of a population of suspended beads in or around an object of interest
3. Analize them with MATLAB to obtain colormaps of relative viscosity

**Discussion points**:

- Are there any blank areas in the map? Why?
- How could we improve spatial coverage?

# Activity 3: Viscoelasticity

**Objective**: we will apply these techniques to a viscoelastic solution to see how they can be used also to characterize complex behaviours.

**Steps**:

1. Prepare a slide as before with  polyethylene oxide (PEO).
2. Analyse them with MATLAB to obtain mean viscosities
3. Compare MSD vs lag curves obtained with Ficoll solutions
4. Estimate viscoelastic properties

**Discussion Points:** Is this setup adequate for unveiling elastic behaviours?